Example 3.45

Graph the equation $y = x + 1$.

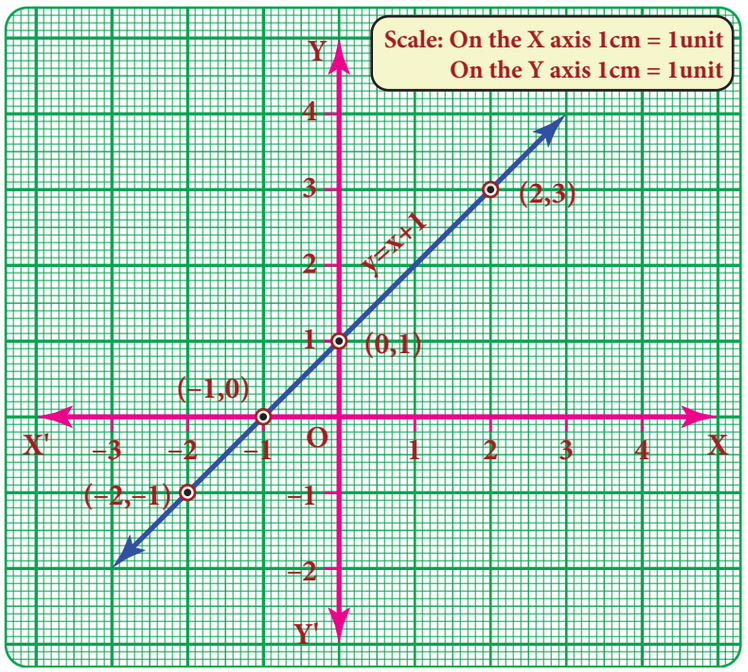
Begin by choosing a couple of values for x and y. It will firstly help to see

(i) what happens to y when x is zero and

(ii) what happens to x when y is zero.

After this we can go on to find one or two more values.

Let us find at least two more ordered pairs. For easy graphing, let us avoid fractional answers. We shall make suitable guesses.



x	−2	−1	0	1	2
y	−1	0	1	2	3

x	$y = x+1$
−2	$y = -2+1 = -1$
−1	$y = -1+1 = 0$
0	$y = 0+1 = 1$
1	$y = 1+1 = 2$
2	$y = 2+1 = 3$

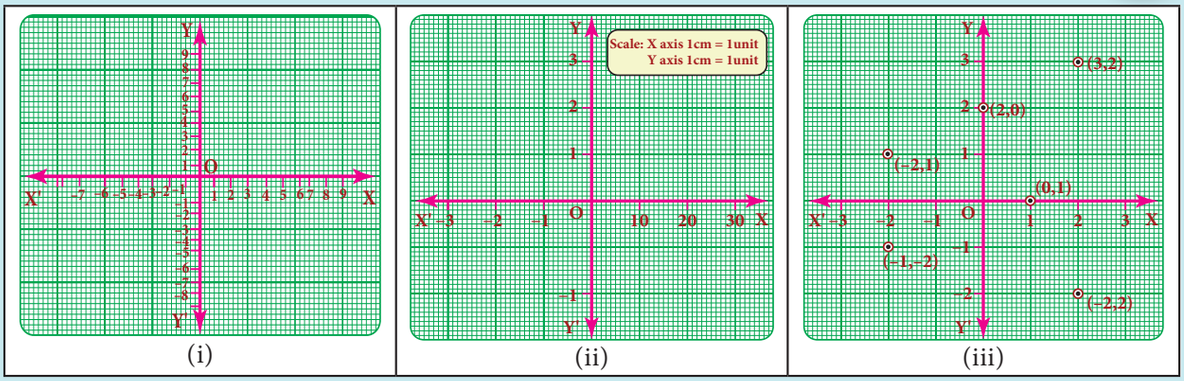
Note

The orientation of the graph will be different according to the scale chosen.

We have now five points of the graph: **(−2,−1) (−1, 0) (0, 1) (1, 2) and (2, 3).**

Try these

Identify and correct the errors



(i) (ii) (iii)

Exercise 3.9

1. Fill in the blanks:

(i) $y = p\,x$ where $p \in z$ always passes through the_____.

(ii) The intersecting point of the line $x = 4$ and $y = -4$ is_____.

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